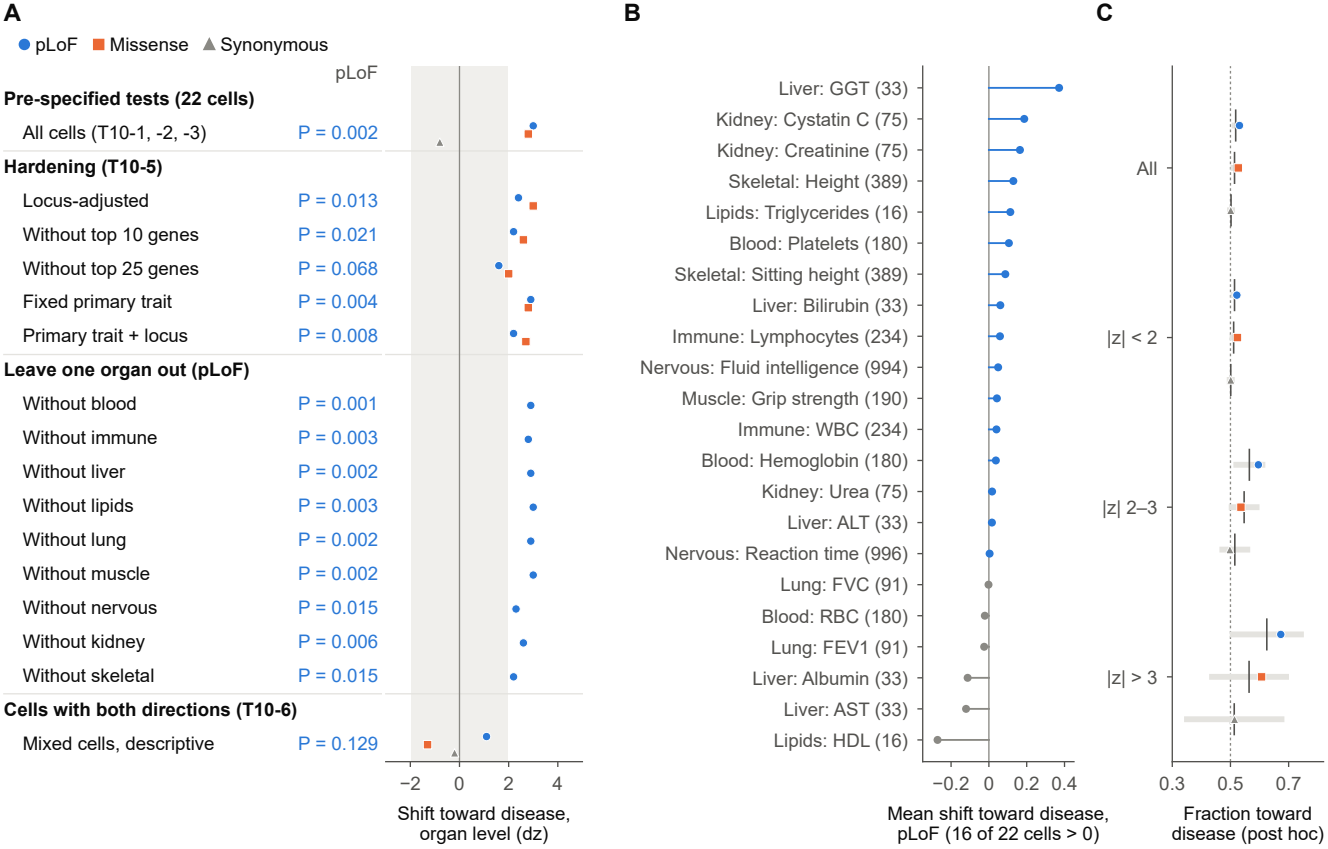

A
pLoF
Missense
Synonymous
pLoF
Pre-specified tests (22 cells)
All cells (T10-1, -2, -3)
P = 0.002
Hardening (T10-5)
Locus-adjusted
P = 0.013
Without top 10 genes
P = 0.021
Without top 25 genes
P = 0.068
Fixed primary trait
P = 0.004
Primary trait + locus
P = 0.008
Leave one organ out (pLoF)
Without blood
P = 0.001
Without immune
P = 0.003
Without liver
P = 0.002
Without lipids
P = 0.003
Without lung
P = 0.002
Without muscle
P = 0.002
Without nervous
P = 0.015
Without kidney
P = 0.006
Without skeletal
P = 0.015
Cells with both directions (T10-6)
Mixed cells, descriptive
P = 0.129
−2
0
2
4
Shift toward disease, organ level (dz)
B
Liver: GGT (33)
Kidney: Cystatin C (75)
Kidney: Creatinine (75)
Skeletal: Height (389)
Lipids: Triglycerides (16)
Blood: Platelets (180)
Skeletal: Sitting height (389)
Liver: Bilirubin (33)
Immune: Lymphocytes (234)
Nervous: Fluid intelligence (994)
Muscle: Grip strength (190)
Immune: WBC (234)
Blood: Hemoglobin (180)
Kidney: Urea (75)
Liver: ALT (33)
Nervous: Reaction time (996)
Lung: FVC (91)
Blood: RBC (180)
Lung: FEV1 (91)
Liver: Albumin (33)
Liver: AST (33)
Lipids: HDL (16)
−0.2
0
0.2
0.4
Mean shift toward disease, pLoF (16 of 22 cells > 0)
C
All
|z| < 2
|z| 2–3
|z| > 3
0.3
0.5
0.7
Fraction toward disease (post hoc)